

Analogue & Digital Communication (EE3003)

Date: September 22, 2025

Course Instructor(s)

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Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 60

Total Questions: 2

Roll No

Section

Student Signature

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1. Attempt all questions and remember to solve parts of the same question together.
2. Show all the steps with the help of diagrams and equations.

CLO # 01: Explain the behavior of a communication system and the noise/distortion therein.

Q1:

[30 marks]

- (a) Draw block diagram of Shannon-Weaver Model for Communication. Discuss in detail the channel impairments caused by *noise* (both external and internal) and *distortion* (both linear and non-linear) with examples. [5]
- (b) Describe the exponential relationship between SNR (Signal-to-Noise ratio) and B (Bandwidth) elaborating the exchange between them for *power-limited* and *band-limited* channels. [5]
- (c) A wireless receiver block is shown in the Fig 1. below. It consists of a receiving antenna which delivers a signal to a bandpass filter. The antenna feed has a loss of -3.6 dB and the filter's loss is -3.1 dB. [4]

$$P_{osc} = +14.2 \text{ dBm}$$

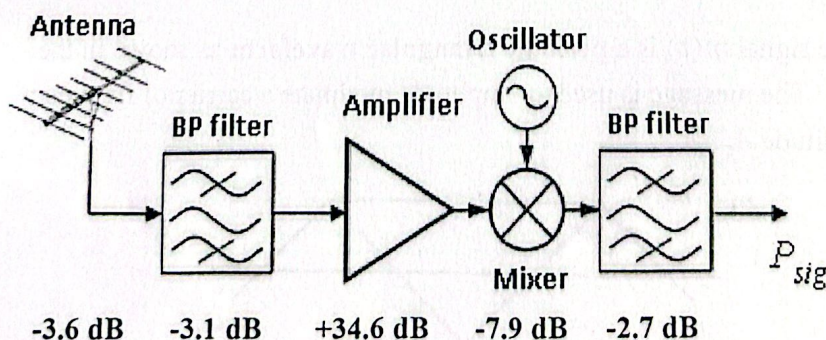


Figure 1. Wireless Receiver with Gain and Loss

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The filter's output drives an amplifier with gain of +34.6 dB and this RF signal is input to a mixer which performs a frequency translation to an IF signal (but the mixer has a loss of -7.9 dB).

Finally, the IF signal travels through another bandpass filter with a nominal loss of -2.7 dB. If the signal strength output by the last filter is $P_{sig} = -90$ dBm, find the signal strength of the signal P_{ant} incident upon the antenna itself and express P_{ant} in milliwatts (mW).

- (d) A communication channel has a signal power $S = 25$ milliwatts (mW) with an average noise power $N = 3$ mW. The channel's transmission medium has a bandwidth B of 2 MHz. [6]

(i) Find the channel capacity. $C = B \log_2 (1 + SNR)$

(ii) Again, compute the channel capacity if the SNR is doubled and the bandwidth B is reduced by a factor of 4.

- (e) A certain channel has ideal amplitude, but non-ideal phase response as shown in the Fig. 2, given by

$$|H(\omega)| = 1$$

$$\theta_h(\omega) = -\omega t_0 - k \sin \omega T, \quad k \ll 1$$

Hint: Use approximation,

$$e^{-jk \sin \omega T} \approx 1 - jk \sin \omega T.$$

- (i) A pulse $g(t)$ bandlimited to B Hz is applied at the input of this filter. Find and sketch the output, $y(t)$.
- (ii) Discuss whether and how this channel will affect TDM and FDM systems from the view point of interference among multiplexed signals.

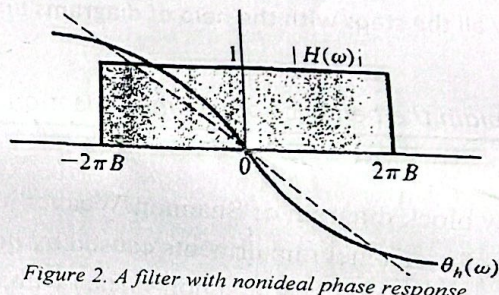


Figure 2. A filter with nonideal phase response

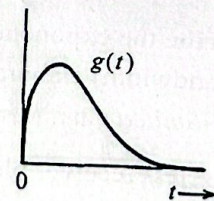


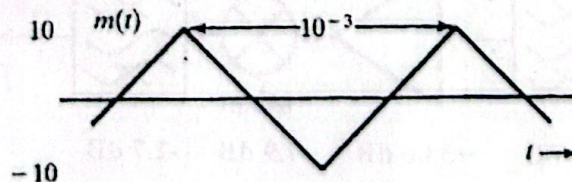
Figure 3. Modulating signal waveform $m(t)$

CLO # 02: Apply the concept of sampling, analog amplitude and frequency modulation and demodulation techniques

Q2:

[30 marks]

- (a) The message signal $m(t)$ is a periodic triangular waveform as shown in the Fig 3. below. The message is used to amplitude modulate a carrier of frequency ω_c and amplitude A_c .



- (i) Sketch the AM signal, $\phi_{AM}(t)$ corresponding to the modulation index: $\mu = 0.5$. [5]
- (ii) Calculate the carrier power P_c , sideband power P_{SB} , and the total transmitted power P_T . [5]
- (iii) Find the power efficiency η of the AM system. [5]
- (b) Fig. 4 shows a switching demodulator circuit used for amplitude demodulation. [15]
Draw the circuit between ab . Mention the type of demodulation and mathematically **exposit** the output expression $z(t)$.

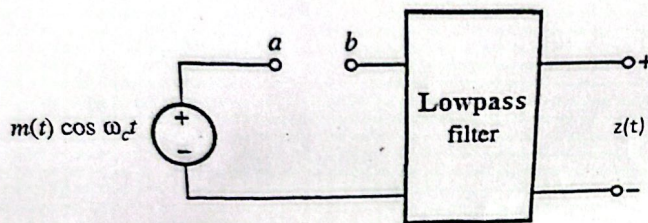


Figure 4. Switching Demodulator Circuit

Hint:

$$w(t) = \frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

$$w_0(t) = \frac{4}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

